

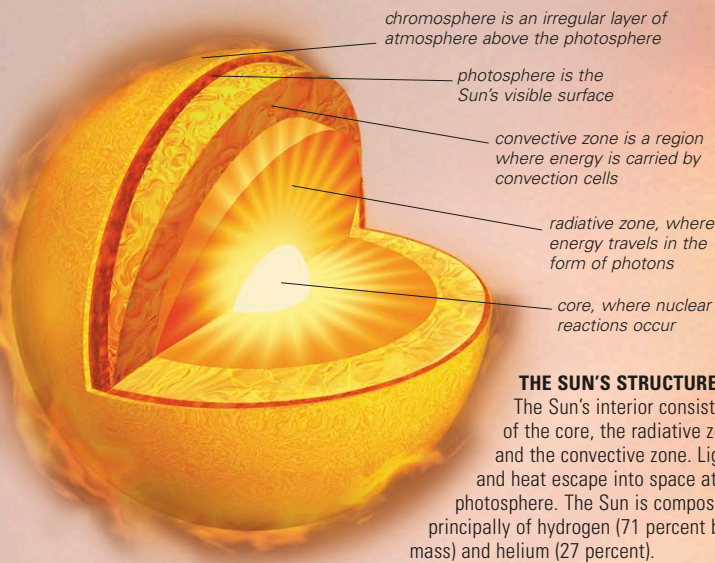
THE SUN

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THE SUN IS A 4.6-BILLION-YEAR-OLD main-sequence star. It is a huge sphere of exceedingly hot plasma (ionized gas) containing 750 times the mass of all the solar system’s planets put together. In its core, nuclear reactions produce helium from hydrogen and generate colossal amounts of energy. This energy is gradually carried outward until it eventually escapes from the Sun’s surface.

INTERNAL STRUCTURE

The Sun has three internal layers, although there are no sharp boundaries between them. At the center is the core, where temperatures and pressures are extremely high. In the core, nuclear fusion turns the nuclei of hydrogen atoms (protons) into helium nuclei at the rate of about 600 million tons per second. Released as byproducts of the process are energy, in the form of photons of electromagnetic (EM) radiation, and neutrinos (particles with no charge and almost no mass). The EM radiation travels out from the core through a slightly cooler region, the radiative zone. It takes about 1 million years to find its way out of this zone, as the photons are continually absorbed and reemitted by ions in the plasma. Farther out, the energy wells up in a convective zone—where huge flows of rising hot plasma occur next to areas of falling cooler plasma—and is transferred to a surface layer called the photosphere. There, it escapes as heat, light, and other forms of radiation.



SUN PROFILE

AVERAGE DISTANCE FROM EARTH 93.0 million miles (149.6 million km)	ROTATION PERIOD (POLAR) 34 Earth days
SURFACE TEMPERATURE 9,932°F (5,500°C)	ROTATION PERIOD (EQUATORIAL) 25 Earth days
CORE TEMPERATURE 27 million °F (15 million °C)	MASS (EARTH = 1) 333,000
DIAMETER AT EQUATOR 864,900 miles (1.4 million km)	SIZE COMPARISON
OBSERVATION The Sun has an apparent magnitude of -26.7 and should never be observed directly with the naked eye or any optical instrument. It can be observed safely only through special solar filters.	

VIOLENT SUN
This composite image taken by the SOHO observatory shows both the Sun's surface and its corona. When the corona image was taken, billions of tons of matter were being blasted through it into space.